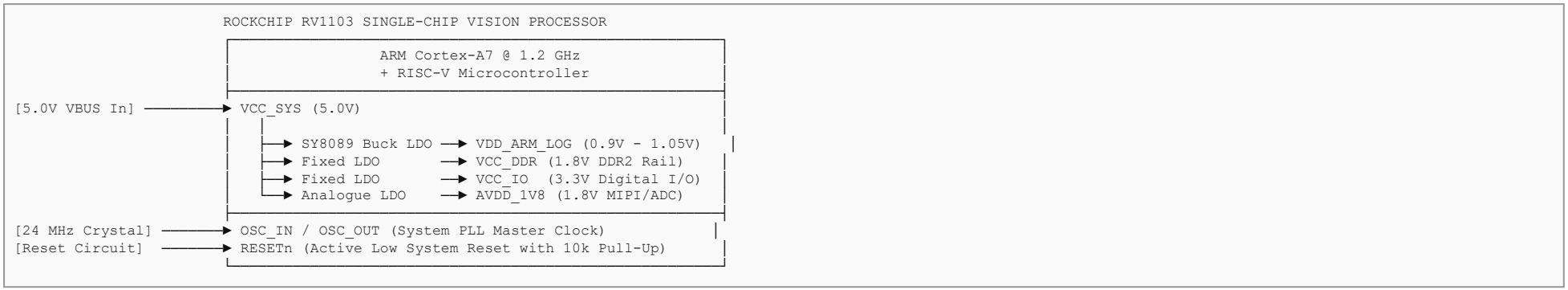
SBC SCHEMATIC SHEET 1: RV1103 CORE, POWER TREE & CLOCKING

1. ROCKCHIP RV1103 CORE PROCESSING & POWER DISTRIBUTION TOPOLOGY



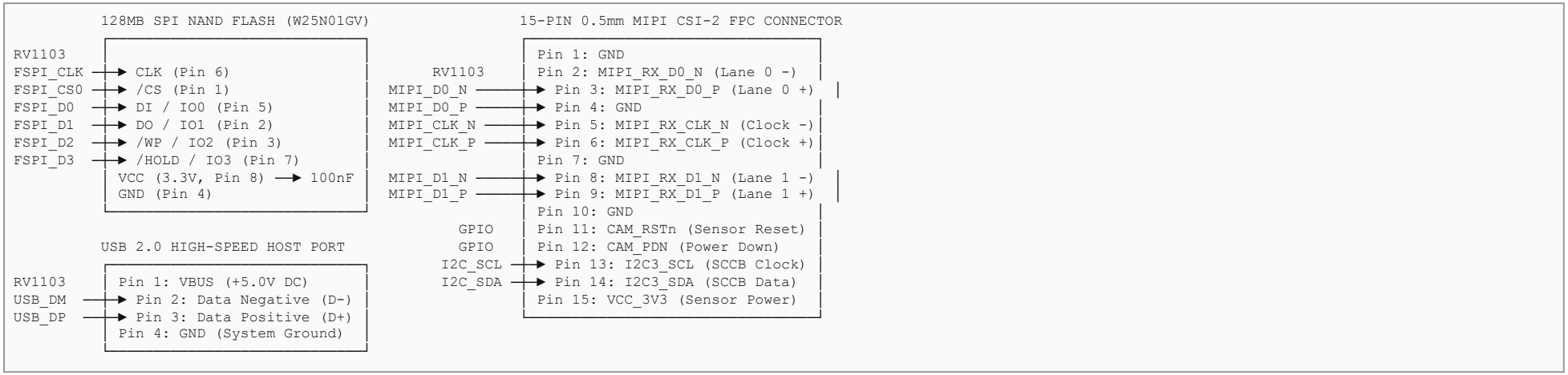
2. POWER TREE VOLTAGE RAIL SCHEDULE

NET NAME	REGULATOR IC	NOMINAL VOLTAGE	TARGET SUBSYSTEM NODE	DECOUPLING CAPACITORS
VCC_SYS	External Input (5.0V Pin)	5.0V DC (4.75V - 5.25V)	Main system power entry rail, USB VBUS input	10uF Tantalum + 100nF Ceramic
VDD_CORE	Internal DC-DC (SY8089)	0.90V - 1.05V DC (DVFS)	ARM Cortex-A7 CPU core, NPU matrix, ISP logic	22uF MLCC + 4x 100nF Ceramic
VCC_DDR	Internal Step-Down LDO	1.8V DC (DDR2 Level)	On-chip 64MB DDR2 memory array & PHY	10uF MLCC + 2x 100nF Ceramic
VCC_IO	Internal Step-Down LDO	3.3V DC (+/- 5%)	GPIO headers, SPI NAND bus, I2C, UART peripherals	10uF MLCC + 100nF Ceramic
AVDD_1V8	Low-Noise Analogue LDO	1.80V DC Filtered	MIPI CSI-2 D-PHY receiver, internal PLL oscillators	4.7uF MLCC + 100nF Ceramic

ORGANIZATION / PROJECT DASHCAM PROTOTYPE PROJECT	SIZE A4	DOCUMENT NUMBER ENG-DASH-SBC-SCH-001	DATE 2026-10-08	SHEET 1 OF 2
		DESIGN ENGINEER Souvik H	ENGINEER REGISTRATION NO. IN-ENG-2022-22824	
DOCUMENT TITLE SBC SCHEMATICS SPECIFICATION	REV 1.4	ENGINEERING DISCIPLINE EMBEDDED HARDWARE & SCHEMATICS	STATUS RELEASED	

SBC SCHEMATIC SHEET 2: 128MB SPI NAND, MIPI CSI & USB INTERFACES

3. 128MB SPI NAND FLASH & MIPI CSI-2 CAMERA CONNECTOR INTERCONNECT



4. BOARD PINOUT & PERIPHERAL MAPPING SUMMARY

HEADER PIN	DEFAULT GPIO	MULTIPLEXED SIGNAL	TARGET FUNCTION IN DASHCAM PROJECT
Pin 1	GPIO1_C7	UART2_TX	Serial Debug Console (115200 / 1.5Mbps)
Pin 2	GPIO1_C6	UART2_RX	Serial Debug Console (Receive)
Pin 5	GPIO1_D2	I2C3_SCL	SC3336 Camera Configuration Clock
Pin 6	GPIO1_D3	I2C3_SDA	SC3336 Camera Configuration Data
Pin 10	GPIO1_B0	GPIO48	Status Optical LED Drive (via 470 Ohm)
Pin 14	GPIO1_A2	GPIO34	Tactile Power / Mode Button (with 10k Pull-Up)
Pin 19, 20	5V_IN, GND	VCC_SYS, GND	Primary System DC Power Connection from PMIC

ORGANIZATION / PROJECT DASHCAM PROTOTYPE PROJECT	SIZE A4	DOCUMENT NUMBER ENG-DASH-SBC-SCH-001	DATE 2026-10-08	SHEET 2 OF 2	
DOCUMENT TITLE SBC SCHEMATICS SPECIFICATION	REV 1.4	DESIGN ENGINEER Souvik H	ENGINEER REGISTRATION NO. IN-ENG-2022-22824		
		ENGINEERING DISCIPLINE EMBEDDED HARDWARE & SCHEMATICS	STATUS RELEASED		